

Supplementary Material

Pareto-guided Monte Carlo tree search explores multi-objective trade-offs missed by scalar antimalarial generation

S1 Pareto-front data provenance

The canonical merged Pareto front (main text, Table 1) comprises four mutually non-dominated candidates. Every candidate is traceable to the deposited per-seed Pareto CSV `p4_pareto_seed_N.csv` from which it originated, and all MPO, SA, RRS and PNS values are identical between the per-seed record and the merged front (`results/pareto/merged_pareto_front.csv`). The only score that differs is SYBA, by design: the original Pareto-MCTS run returned a constant zero fallback for every molecule (SYBA was inactive during search, as recorded in the per-seed CSVs), so SYBA was recomputed post-hoc with the lich/conda classifier and the recomputed values were used to re-derive the non-dominated front and its hypervolume. This procedure is documented in the main-text Methods and is fully reproducible: the script `scripts/p4_recompute_pareto_syba.py` regenerates the merged CSV and the log `merged_pareto_front_recompute.log` records the raw and normalised SYBA scores (78.67, 58.13, -19.25, 25.88) and the hypervolume (1.2366). The verification script `scripts/p4_pareto_provenance_check.py` re-runs all of these checks and exits with status 0 only when every SMILES and score in the merged front matches a deposited record.

[table 1](#) lists the four canonical candidates with their provenance.

Table 1: Canonical merged Pareto front with per-seed provenance. All four points are mutually non-dominated on the active objectives (MPO, SYBA, RRS, PNS; SA is constant at 3.0 and therefore excluded from the hypervolume). SYBA was recomputed post-hoc (main text, Methods); MPO, SA, RRS and PNS are unchanged from the per-seed records.

Candidate (SMILES)	MPO	SYBA	SA	RRS	PNS	seed	origin CSV
<chem>CNC(=O)c1ccc(OC(C)C(C)(C)C)c(OC)c1</chem>	0.910	1.000	3.0	0.211	1.0	13	<code>p4_pareto_seed_13.csv</code>
<chem>COc1c(C)cc(-c2ccc(S(N)(=O)=O)cc2)c1C</chem>	0.945	1.000	3.0	0.196	1.0	18	<code>p4_pareto_seed_18.csv</code>
<chem>CC1CCC(n2cccn2)NC1(c1ccccc1)C1OCCO1</chem>	0.946	0.021	3.0	0.141	0.0	5	<code>p4_pareto_seed_5.csv</code>
<chem>COc1cc(CO)c(O)c(C)c1C</chem>	0.729	0.994	3.0	0.223	1.0	6	<code>p4_pareto_seed_6.csv</code>

S2 Molecular diversity of the four generation methods

The main text reports scalar-reward statistics only, because the primary contributions are the honest benchmark ranking and the multi-objective Pareto front. Molecular diversity is analysed here in the Supplementary Material using real, deposited molecule sets.

Molecule sets. For each of the four generation methods (MCTS+ScafVAE, random search, greedy search, genetic algorithm) the best-scoring molecule of each of the 20 independent benchmark seeds was recorded, giving a deposited set of 20 molecules per method (`results/benchmark_molecules/p4_benchmark_molecules_seed_N.csv`). These are the same runs whose scalar rewards are reported in the main-text benchmark table ([Table 3](#)). The benchmark was re-run for this submission (2026-08-02) with the committed script `p4_mcts_benchmark.py`, a value-preserving vectorisation of the docking Tanimoto scan, and the screened-optimal MCTS configuration ($c_{\text{PUCT}} = 5.0$, $\nu = 0.01$, $T = 0.8$, ScafVAE policy wired into PUCT priors; main text, Hyperparameter screening); all 20 per-seed runs reproduce deterministically from the deposited code and data (`results/benchmark\_molecules\_opt/`). This canonical run supersedes both the v9 deposit (2026-07-29), whose rewards could not be reproduced from the committed code because the docking oracle’s Tartarus library state differed at run time, and an interim default-configuration re-run; the method ranking

(Random > MCTS > GA > Greedy) is identical across all three, and the deposited molecule sets are the canonical outputs of the benchmark.

Descriptors. ECFP4 fingerprints (Morgan radius 2, 2048 bits) were computed with RDKit. Pairwise Tanimoto dissimilarity is $1 - \text{Tanimoto}$; the reported value is the mean over all unique pairs (190 pairs per method). Bemis–Murcko scaffolds were extracted with RDKit’s `MurckoScaffoldSmiles`; the unique-scaffold fraction is the number of distinct scaffolds divided by the number of valid molecules. Two-dimensional MDS coordinates (metric MDS on the Tanimoto dissimilarity matrix) are deposited for visualisation (`results/diversity/p4_diversity_mds.csv`).

[table 2](#) reports the real diversity statistics.

Table 2: Real molecular diversity of the four generation methods computed from the deposited molecule sets (20 best-in-seed molecules per method). Values are computed with RDKit; deposited CSVs: `p4_diversity_metrics.csv` and `p4_benchmark_molecules_*`.

Method	n	Mean pairwise	Unique	Unique scaffold	
		Tanimoto dissimilarity	Bemis–Murcko scaffolds	fraction	
MCTS+ScafVAE	20	0.802 3	19	0.950	Mean pairwise
Random	20	0.780 6	20	1.000	
Greedy	20	0.000 0	1	0.050	
GA	20	0.833 2	17	0.850	

dissimilarity is the mean $1 - \text{Tanimoto}$ over all unique pairs (190 pairs per method); scaffolds are unique Bemis–Murcko skeletons (RDKit); the fraction is unique scaffolds divided by valid molecules.

Table 3: Canonical re-benchmarked scalar rewards (20 seeds; mean, standard deviation, min, max). MCTS uses the screened optimal configuration (`c_puct=5.0`, `virtual_loss=0.01`, `T=0.8`, ScafVAE policy wired). Deposited CSV: `p4_benchmark_merged.csv` under `results/benchmark_molecules_opt/`.

Method	Mean reward	Std	Min	Max	Mean time (s)
Random	0.733 5	0.006 3	0.722 7	0.748 1	42.2
MCTS+ScafVAE	0.727 6	0.009 0	0.705 4	0.744 2	82.2
GA	0.702 7	0.015 2	0.674 9	0.731 1	1.9
Greedy	0.614 7	0.000 0	0.614 7	0.614 7	0.1

Because greedy search collapses deterministically to a single local optimum (std identically zero in the canonical benchmark), its 20 per-seed outputs coincide, yielding zero pairwise diversity and a single unique Bemis–Murcko scaffold by construction (Table 2). The three stochastic methods generate structurally distinct outputs per seed: random search achieves full scaffold coverage (20 unique scaffolds from 20 molecules), while MCTS and GA return 19 and 17 unique scaffolds respectively. Mean pairwise Tanimoto dissimilarity is high (0.78–0.83) for all three stochastic methods, indicating that seeded exploration remains structurally diverse despite the curated fragment vocabulary.

S3 Reproducibility

All deposited CSVs, the benchmark, Pareto and diversity analysis scripts, and the provenance-check script are listed in the Zenodo manifest. The canonical benchmark was re-run with the exact script `p4_mcts_benchmark.py` (with the addition of best-SMILES recording, which does not alter the reward semantics) and a value-preserving vectorisation of the docking Tanimoto scan (verified to return identical similarities and indices on held-out molecules); the 20 per-seed rewards in `results/benchmark_molecules/p4_benchmark_merged.csv` are reproduced deterministically from the deposited code and data. The diversity statistics are computed by `scripts/p4_compute_diversity.py` from the deposited molecule sets, and the provenance of the Pareto front is verified by `scripts/p4_pareto_provenance_check.py`.